Integrated Automated Feeding and Watering System with Centralized Data Monitoring for Commercial Piggeries

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***Abstract*—Manual feeding and inadequate health monitoring in commercial piggeries contribute to feed waste, undetected illness, and preventable livestock losses. This paper presents the design and development of an integrated, hardwired automated feeding and watering system with centralized data monitoring for commercial piggeries. The system employs a main control module connected to multiple feeder modules via a wired network, integrating weight sensors, water flow sensors, and air quality sensors to ensure precise feed rationing and continuous environmental monitoring. Collected data is processed and displayed through a local user interface, enabling caretakers to track consumption patterns and detect early signs of illness in real time. By eliminating reliance on manual observation and wireless communication, the system offers a reliable and scalable solution for improving biosecurity, animal health management, and operational efficiency in swine production.**

***Index Terms*—Livestock Automation, Precision Feeding, Biosecurity, Real-time Monitoring**

I. Introduction

Livestock farming is a fundamental cornerstone of global agriculture, food security, and economic stability. As the global population surpasses 8.2 billion in 2025, livestock farming continues to play a vital role in food security, rural livelihoods, and international trade [?] [?]. Livestock is also a major economic driver. The gross value of global livestock production is projected to increase by 14% over the next decade, with the livestock sector leading this expansion with a 16% rise [?]. Global dietary preferences are also shifting toward nutrient-rich animal proteins due to rising incomes and urbanization, especially in middle-income countries in Asia and Latin America [?] [?]. By 2034, daily per capita consumption of fish and livestock products is expected to increase by 25% in lower-middle-income nations [?] [?].

Swine production is one of the most important sources of animal protein in the world and is heavily regionally concentrated within the larger livestock industry [?]. Pork is the second most-produced meat after poultry, with an estimated production of 116.27 million metric tons in 2025 [?]. China remains the leading producer, with an inventory of approximately 400 million pigs and about 49% of total global pork production. The European Union (19%) and the United States (11%) are the next largest producers [?]. Swine farming is also a crucial component of the cultural and economic systems of many developing nations. In the Philippines, for example, the swine industry is valued at around ₱191 billion and accounts for 78% of the country's total livestock output [?] [?] [?]. Around 71.1% of the Philippine hog inventory is managed by backyard smallholder farmers, highlighting the importance of piggery farming to local economic survival [?].

The global swine industry is highly susceptible to transboundary diseases, most notably **African Swine Fever (ASF)**, which has become a major threat to swine-sector growth [?]. In this context, pig health monitoring is essential for early disease detection, economic sustainability, improved animal welfare, and reduced dependence on antibiotics in agriculture. Although manual inspection of pens, pig behavior, and air quality remains common practice, modern swine farming is steadily shifting toward advanced, continuous monitoring technologies to achieve these goals [?]. Since the first confirmed ASF outbreak in the Philippines in July 2019, the disease has wiped out an estimated 6 million pigs, causing the country's swine population to fall to its lowest level in years [?] [?]. By 2023, ASF had already caused an estimated ₱100 billion in economic losses [?]. As local pork production declined sharply, the country became increasingly dependent on meat imports, which reached an estimated 1.64 million metric tons in 2025 [?] [?]. This dependence on imported pork, combined with supply shortages, contributed to a significant increase in pork prices nationwide.

In response, both public and private sectors have invested in health monitoring and early detection systems to stabilize the industry and limit disease spread. The Department of Agriculture (DA) launched the Bantay ASF sa Barangay (BABay ASF) program to strengthen local monitoring and institutionalize biosecurity practices [?] [?]. In addition, DOST-PCAARRD funded the BRIDGES project, which established mobile laboratory units and uses advanced pathogen-detection technologies to provide rapid, on-site disease diagnosis in rural and resource-limited areas [?]. Manual pig monitoring is highly time-consuming and prone to error, especially when caregivers are overworked. In a case study conducted in a piggery in Tuburan, Cebu, the inability of caretakers to constantly monitor every pig led to the death of 20 nursery pigs in a single month [?]. Meanwhile, commercial operations in regions such as Central Luzon and CALABARZON are increasingly adopting climate-controlled, tunnel-ventilated housing with digital sensors as the sector modernizes. These monitoring conditions have helped reduce pre-weaning mortality from 10% to 9.43% while improving productivity and reproductive performance [?] [?].

II. Statement of the Problem

Inconsistent feeding and drinking practices in pigs can lead to significant health risks and feed wastage [?]. Early detection of illness through physical or manual observation is often unreliable because subtle symptoms may go unnoticed [?]. Without a consistent tracking system, caretakers must rely on labor-dependent observation, which can result in inaccurate records and delays in identifying symptoms or other problems. A pig's health status is closely linked to its feeding and drinking behavior, as measurable deviations in dietary intake are often among the first observable indicators of illness [?]. However, without a data-recording system, individual feeding patterns are not established or tracked over time, making it difficult to distinguish normal behavioral variation from significant decline. In addition, unmonitored air quality in piggery environments poses an unnecessary and preventable risk of common respiratory illnesses such as coughs and colds among pigs [?].

This study addresses these issues by proposing a system that establishes per-module consumption baselines using a rolling time window and flags deviations beyond a defined threshold as potential signs of abnormal behavior requiring immediate caretaker attention. Air-quality readings from integrated sensors are also compared against safe-range standards, with alerts generated when toxicity levels within the pig-pen environment exceed acceptable limits. This approach aims to reduce reliance on subjective visual observation and provide a more consistent and reliable basis for early health monitoring.

III. Goals and Objectives

The main objective of this project is to design and develop an integrated automated swine feeding and watering system with centralized data monitoring and control.

To achieve this goal, the system must fulfill the following specific objectives:

- Implement a programmable central unit to manage main storage and automate the dispensing process.
- Design individual feeder modules capable of monitoring feed consumption, water availability, and ammonia levels.
- Develop a user interface on the central unit for system monitoring, data acquisition, and historical logging.
- Integrate an SMS module to alert caretakers when the system detects consumption or environmental anomalies.

IV. Scope and Limitations

This study focuses on the design and development of an automated feeding and watering system with data monitoring for commercial piggeries. The system is intended to support swine management by combining scheduled feed and water dispensing, sensor-based environmental monitoring, and digital record-keeping within a farrowing piggery prototype setup.

This study is limited to the following:

- The system does not perform direct veterinary diagnosis or laboratory disease confirmation.
- Implementation is limited to the system configuration used, with dimensions based on the farrowing pen setup; performance may vary across different farm layouts.
- System operation may be affected by interruptions in electrical power.
- Air quality monitoring is limited to sensors integrated within the feeder modules and covers only the immediate environment of each pen, not the entire piggery facility.

V. Significance of the Study

This study holds significant value for multiple sectors, ranging from small-scale backyard farmers to major commercial piggery operators, as well as broader agricultural and technological stakeholders.

Swine Farmers and Caretakers

By replacing labor-intensive, error-prone manual feeding and monitoring procedures with an automated, data-driven approach, the proposed solution directly addresses the operational strain on caregivers. As shown in the Tuburan, Cebu case study, where 20 nursery pigs died in a single month due to insufficient attention and monitoring, the system helps minimize feed waste and enables caretakers to identify early signs of illness before they escalate into costly or fatal outcomes through precise feed rationing and continuous tracking of consumption behavior.

Animal Health and Biosecurity

Early detection of health abnormalities is more critical than ever because of the severe effects of African Swine Fever on the Philippine swine industry. By continuously logging feed intake, water consumption, and air-quality data, the proposed system establishes a structured, timestamped record of each pig pen's normal operating conditions. Deviations from these baselines are flagged as alerts for caretaker follow-up. This rule-based approach offers a non-invasive, data-driven layer of biosecurity support that operates independently of internet connectivity. The system is designed to assist in the early detection of possible health concerns in pigs, but it does not perform actual clinical diagnosis or disease confirmation. Planned validation against documented health events will be used to characterize the system's detection reliability in terms of alert accuracy and response timeliness. The system also complements existing government biosecurity programs such as BABay ASF by contributing structured farm-level data to support broader disease-monitoring efforts.

The Philippine Swine Industry

With approximately 71.1% of the Philippine hog inventory managed by smallholder backyard farmers, the local swine sector remains highly vulnerable to disruptions caused by disease and inefficient farm management. A reliable, centralized automated system could help stabilize domestic pork production, reduce dependence on imported meat, which reached a record 1.64 million metric tons in 2025, and support the national goal of rebuilding the country's swine population to pre-ASF levels.

Field of Agricultural Automation and Embedded Systems

This study contributes to the growing body of research on IoT and precision agriculture applied to livestock management. By demonstrating a fully hardwired, sensor-integrated feeding and monitoring system designed specifically for the Philippine piggery context, the project provides a practical reference for future engineering work in agricultural automation, particularly in environments where wireless reliability may be a concern.

Future Researchers

The system's data logging and analytics capabilities generate structured, timestamped records of feed consumption, water intake, and environmental conditions. This dataset can serve as a foundation for future studies on swine behavioral patterns, machine learning-based health prediction models, and the broader integration of precision livestock farming technologies in developing agricultural economies.

VI. Review of Related Literature

A. Automated Feeding and Watering Systems

Automated feeding and watering systems in commercial piggeries are designed to deliver controlled and consistent amounts of feed and water using sensor-based monitoring and programmable control logic. These systems aim to reduce feed wastage and improve efficiency by replacing manual, labor-dependent practices with precise and repeatable operations [?], [?].

Literature emphasizes that inconsistencies in feeding and drinking behavior can lead to health deterioration and inefficient resource utilization [?], [?]. As a result, automated systems are designed to standardize feeding schedules and quantities while ensuring that consumption is monitored continuously. A key consideration in system design is maintaining consistency across multiple feeding points, which requires coordinated control and reliable data exchange within the system.

B. Health Monitoring Through Behavioral and Consumption Data

Behavioral monitoring through feeding and drinking patterns is widely recognized as a non-invasive method for early disease detection in swine. Changes in consumption behavior often occur before visible clinical symptoms, making them valuable indicators of potential health issues [?], [?].

Studies such as Dingcong et al. demonstrate that tracking feeder and drinker interactions can provide insight into animal health and behavior [?]. More broadly, research highlights the importance of establishing baseline consumption patterns and identifying deviations over time [?], [?]. These deviations can be analyzed using time-series approaches, where historical data is used to define normal behavior and detect anomalies.

Without structured data collection and analysis, caretakers rely on subjective observation, which may result in delayed detection of illness. The literature supports the use of continuous monitoring systems to provide objective and data-driven assessment of animal health conditions [?], [?].

C. Air Quality Sensing in Swine Environments

Environmental conditions within piggery enclosures play a significant role in animal health and productivity. Poor air quality has been associated with respiratory illnesses and reduced overall welfare in swine populations [?], [?].

Research indicates that monitoring environmental parameters enables early identification of unsafe conditions and supports timely intervention. Air quality data also complements behavioral monitoring, as environmental stressors can influence feeding and drinking patterns [?], [?]. This relationship highlights the importance of integrating environmental and behavioral data to achieve a more comprehensive understanding of animal health.

Existing studies also explore advanced approaches such as sound-based detection of respiratory symptoms, further reinforcing the role of non-invasive monitoring techniques in livestock management [?].

D. Centralized Data Monitoring and User Interface Design

Centralized monitoring systems are widely used in agricultural automation to aggregate data from multiple sensing points and provide a unified view of system status. These systems enable real-time monitoring, data logging, and alert generation, supporting informed decision-making by farm operators [?], [?].

The literature emphasizes the importance of accessible and intuitive user interfaces that present complex data in a simplified manner. Features such as real-time dashboards, historical trend visualization, and threshold-based alerts are commonly identified as essential components of effective monitoring systems [?].

In addition, centralized systems reduce reliance on manual observation and improve the accuracy and consistency of recorded data. Continuous monitoring technologies have been shown to enhance productivity and reduce labor demands in modern swine farming operations [?], [?].

E. African Swine Fever and Technology-Driven Biosecurity

The impact of African Swine Fever (ASF) on the global and Philippine swine industry has underscored the importance of early detection and preventive monitoring systems. The disease has caused significant livestock losses and economic disruption, prompting increased investment in biosecurity and surveillance initiatives [?], [?].

Programs such as the Bantay ASF sa Barangay (BABay ASF) and the BRIDGES project focus on monitoring and rapid diagnosis at the community and national levels [?], [?]. However, literature suggests that farm-level monitoring systems can complement these efforts by providing continuous observation of animal behavior and environmental conditions.

By detecting deviations that may indicate early-stage illness, monitoring systems contribute to preventive health management. This approach aligns with broader efforts to reduce reliance on antibiotics and improve sustainability in livestock production [?].

F. Scalability and Deployment Considerations in Philippine Piggeries

The structure of the Philippine swine industry, characterized by a combination of commercial farms and smallholder operations, presents unique challenges for technology adoption. With a large proportion of hog production managed by backyard farmers, systems must be adaptable, scalable, and accessible [?].

Literature highlights the need for flexible system designs that can accommodate varying farm sizes and operational conditions. Continuous monitoring technologies are increasingly being adopted in modern farming environments, demonstrating their potential to improve productivity and animal welfare [?], [?].

Additionally, non-invasive monitoring approaches, including sound-based and image-based techniques, provide scalable methods for observing animal health across large populations [?], [?]. These approaches expand the range of available tools for precision livestock farming and support the gradual modernization of the sector.

VII. Methodology

This chapter presents the conceptual framework underpinning the system's design philosophy, followed by a detailed account of the system architecture, hardware construction, volume calculations, dispensing mechanism, sensing strategy, communication protocol, and evaluation metrics.

A. Conceptual Framework

The system is conceptualized as a closed-loop precision livestock management platform grounded in two theoretical underpinnings: Precision Agriculture Theory—which frames sensor-driven, individualized resource delivery as the basis

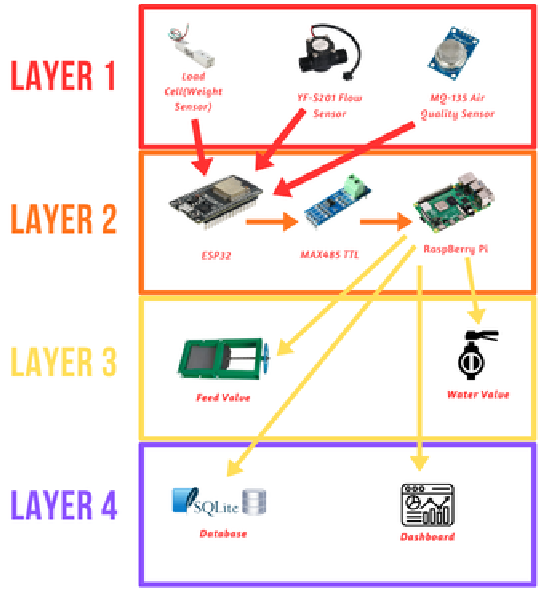

Fig. 1. Conceptual Framework

for efficient swine production—and Feedback Control Systems Theory, which models each feeder module as a node whose output (feed and water dispensed) is continuously logged and compared against defined consumption baselines to flag deviations indicative of illness.

The system's operational logic spans four functional layers:

- **Layer 1 – Sensing (Input):** Each feeder module integrates a load cell with an HX711 24-bit ADC for gravimetric feed measurement, a waterproof ultrasonic sensor (e.g., JSN-SR04T) for water level monitoring, and an MQ-135 electrochemical sensor for ammonia air-quality monitoring.
- **Layer 2 – Processing and Decision (Control):** Each feeder module's ESP32 microcontroller evaluates sensor readings against pre-established rolling-window consumption baselines. Deviations beyond a defined threshold are flagged as health alerts and relayed to the main control module via the RS-485 wired bus. The Raspberry Pi central controller processes incoming data, executes alert logic, and manages the touchscreen dashboard.
- **Layer 3 – Actuation (Output):** Upon a scheduled or manually triggered feeding event, the main control module commands the solenoid-actuated sliding plate valve at the base of the truncated-pyramid hopper to open for a calibrated duration, dispensing a gravimetrically verified quantity of feed into each trough. Water is dispensed through a solenoid valve when the sensor detects the water level has dropped below a predefined threshold. This valve can also be manually overridden using an active-high push button on pin RA0 to ensure rapid caretaker intervention if needed.
- **Layer 4 – Feedback, Logging, and Remote Notification:** All sensor readings, dispensing events, and alerts are timestamped and stored in a local SQLite database

on the Raspberry Pi. In addition to the local touchscreen dashboard, the system utilizes a LoRa (Long Range) transceiver to broadcast critical health and environmental alerts. These alerts are received by a remote standalone terminal located in the caretaker's office, ensuring real-time notification across the farm property without reliance on cellular networks or internet connectivity.

B. System Architecture: Distributed Software Workflow

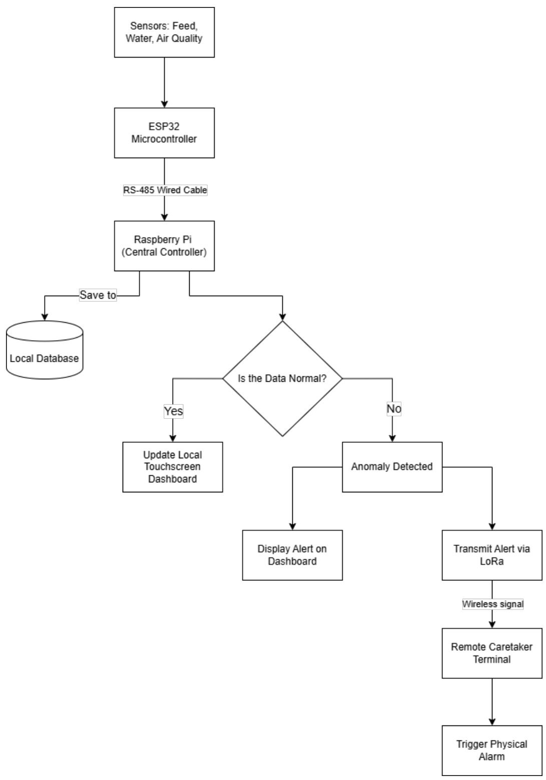

Fig. 2. System Flowchart

The system employs a distributed, hierarchical software architecture to decouple real-time hardware interfacing from high-level data orchestration. As illustrated in Figure **??**, the software workflow operates in a continuous loop. Data acquisition begins at the edge nodes where the ESP32 samples the sensors via hardware interrupts. The Raspberry Pi systematically polls these nodes over the RS-485 bus. Upon receiving the telemetry, the Python backend writes the raw data to the SQLite database and passes it to the Analytical Engine. If the data falls within normal baseline parameters, it is pushed to the Flask frontend for live dashboard visualization. If an anomaly is detected, the system simultaneously triggers a visual alert on the local user interface and dispatches a lightweight alert payload via the Telemetry Broadcaster Service to the remote LoRa terminal.

- **Edge Intelligence (Feeder Nodes):** Each node is governed by an ESP32 dual-core microcontroller. One core is dedicated to continuous sensor polling (load cell and flow meter) using hardware interrupts and the HX711 24-bit ADC, while the second core manages a MODBUS RTU slave stack. Telemetry is stored in a local register map, allowing the master to retrieve data asynchronously without blocking sensor acquisition.
- **Central Orchestration (Main Module):** A Raspberry Pi serves as the MODBUS RTU master, implementing a sequential polling loop via the RS-485 bus. The main module is co-located with the central hopper—a 0.18 m^3-capacity truncated-pyramid storage unit—from which feed is gravity-fed through an auger conveyor to individual feeder nodes. The master logic is divided into four primary services:
  - **Poller Service:** Executes MODBUS read commands at 1-second intervals to aggregate per-pen telemetry.
  - **Analytical Engine:** Implements the rolling-window anomaly detection logic, comparing incoming data against the SQLite-persisted baseline.
  - **Command Service:** Dispatches metered dispensing instructions based on the pre-configured feeding schedule or manual overrides.
  - **Telemetry Broadcaster Service:** Interfaces with a connected LoRa module via SPI or UART. When the Analytical Engine flags an anomaly based on the rolling-window baseline, this service encodes the alert data into a lightweight payload and transmits it over the LoRa network.
- **Remote Client Terminal:** A secondary edge device positioned outside the immediate piggery environment. It decodes incoming LoRa payloads and triggers local physical alarms to immediately notify the caretakers of the detected anomaly.

C. Software Architecture and Technology Stack

The system's software architecture is distributed across the edge feeder nodes and the central control module. The technology stack was deliberately selected to prioritize fault tolerance, low-latency execution, and complete offline capability, ensuring the system remains fully operational without external internet dependencies.

1. Edge Node Firmware (ESP32)

- **Software/Language:** C/C++ utilizing the FreeRTOS (Real-Time Operating System) environment.
- **Why it is used:** C/C++ provides the low-level memory management and fast execution speeds required for high-frequency sensor sampling. FreeRTOS allows the ESP32 to effectively utilize its dual-core processor by dividing tasks into concurrent threads.
- **How it is implemented:** Core 0 is dedicated strictly to hardware interrupts—specifically counting pulses from the YF-S201 flow meter and reading the HX711 ADC—ensuring no data is lost during execution delays. Core 1 manages the MODBUS RTU slave stack, listening for polling requests from the Raspberry Pi and transmitting the localized sensor data stored in its memory registers.

2. Central Controller Backend (Raspberry Pi)

- **Software/Language:** Python 3, Flask Web Framework, and PyModbus.

- **Why it is used:** Python offers extensive library support for serial communication and data analysis. Flask is a lightweight WSGI web application framework that does not require heavy dependencies, making it ideal for an embedded Linux environment. PyModbus provides a highly stable, synchronous protocol for RS-485 communication.
- **How it is implemented:** Python acts as the central orchestrator. A background script runs continuous PyModbus loops to query each ESP32 node. The retrieved data is passed through the Python-based Analytical Engine to calculate rolling averages and detect anomalies. Simultaneously, Flask serves the backend API, exposing this data to the local network so the frontend can retrieve it dynamically without requiring a page refresh.

3. Data Persistence (Database)

- **Software:** SQLite.
- **Why it is used:** Unlike heavy relational databases (e.g., MySQL or PostgreSQL) that require dedicated server processes, SQLite is a C-language library that implements a small, fast, self-contained, high-reliability SQL database engine. It writes directly to ordinary disk files, minimizing CPU and RAM overhead on the Raspberry Pi.
- **How it is implemented:** The Python backend executes standard SQL commands to log timestamped telemetry data, system alerts, and feeding schedules. To prevent SD card degradation from excessive write cycles, database commits are batched and executed at defined intervals rather than instantaneously per sensor read.

4. User Interface and Visualization (Frontend)

- **Software/Language:** HTML5, CSS3, vanilla JavaScript, and Chart.js.
- **Why it is used:** This combination ensures a responsive, highly intuitive dashboard. Chart.js is specifically utilized because it renders lightweight, HTML5 canvas-based time-series graphs that do not lag on embedded processors.
- **How it is implemented:** The Raspberry Pi boots directly into a Chromium browser in "Kiosk Mode" (fullscreen, no navigation bars), pointing to the local Flask server address. JavaScript asynchronous requests (AJAX) continuously pull the latest SQLite data via Flask API endpoints, dynamically updating the Chart.js line graphs and numeric alert gauges so caretakers see real-time shifts in consumption and air quality.

D. Volume and Structural Calculations

The physical dimensions of the central hopper and feeder trough were derived from pig-feed bulk density and a total required storage capacity of 90 kg, using a bulk density of 550 kg/m^3. The minimum required storage volume is:

$$V = \frac{m}{\rho} = \frac{90 \text{ kg}}{550 \text{ kg/m}^3} \approx 0.1636 \text{ m}^3 \tag{1}$$

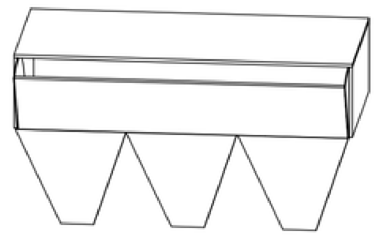

Fig. 3. Main Control Module

An allowance of 0.180 m^3 was adopted as the design target. The central storage hopper is designed as a truncated rectangular pyramid (frustum) with a top face of $L = 1.0$ m $\times W = 0.5$ m and a bottom opening of $a = b = 0.1$ m, consistent with the sliding-plate valve aperture. The slant height of the frustum walls is calculated from:

$$h = \frac{(0.5 \text{ m} - 0.1 \text{ m})}{2} \times \tan(60^\circ) \approx 0.35 \text{ m} \tag{2}$$

The volume of each conical frustum section is given by the prismatoid formula:

$$V_{cone} = \frac{h}{3}(A_B + a_b + \sqrt{A_B \cdot a_b}) \tag{3}$$

where $A_B = 0.5$ m $\times$ 0.333 m $= 0.165$ m^2 is the top-face area per section, and $a_b = 0.1$ m $\times$ 0.1 m $= 0.01$ m^2 is the bottom-aperture area. Substituting into (3):

$$V_{cone} \approx 0.02516 \text{ m}^3 \times 3 \approx 0.075 \text{ m}^3 \tag{4}$$

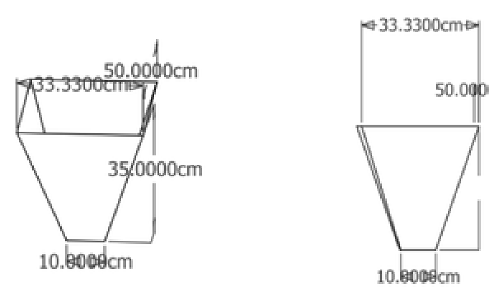

Fig. 4. Cone Dimensions

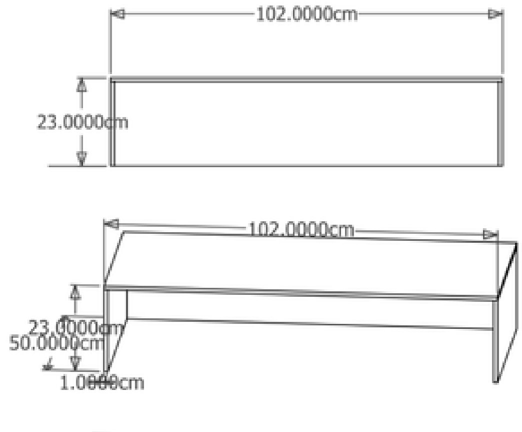

Fig. 5. Box Dimensions

The remaining volume is provided by a rectangular upper reservoir section of height h_{rec}, computed from:

$$V_{rec} = V_{total} - V_{cones} = 0.180 - 0.075 = 0.105 \text{ m}^3 \quad (5)$$

Solving for the rectangular section height with $L = 1.0$ m and $W = 0.5$ m:

$$h_{rec} = \frac{0.105 \text{ m}^3}{1.0 \text{ m} \times 0.5 \text{ m}} = 0.21 \text{ m} \quad (6)$$

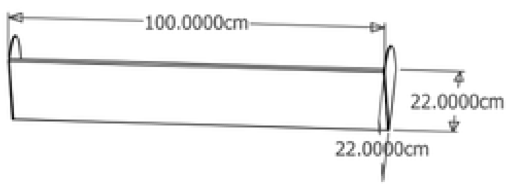

Fig. 6. Hopper Lid Dimensions

The total hopper height is therefore $h_{cone} + h_{rec} = 0.35 + 0.21 = 0.56$ m, housed within an overall footprint of 102 cm $\times$ 50 cm at the top rim. A separate funnel outlet is provided for each feeder node, directing feed into the trough.

E. Dispensing Mechanism

Fig. 7. NEMA 17 Stepper Motor

Feed release from the central hopper is controlled by a solenoid-actuated sliding plate valve (guillotine valve) positioned at the 0.1 m $\times$ 0.1 m bottom aperture of the truncated pyramid. A 12 V DC push-pull solenoid drives the plate linearly, opening or closing the aperture under microcontroller command. This valve type was selected for its simplicity, low power consumption, and compatibility with granular feed materials that would jam rotary or butterfly valves.

An auger screw conveyor driven by a NEMA 17 stepper motor supplements the sliding valve for metered dispensing. By counting motor steps, the ESP32 can dispense feed in calibrated increments independently of gravity head pressure, improving dose accuracy across varying hopper fill levels. The hinged hopper lid is actuated by a secondary servo motor for automated refill access.

F. Sensing Strategy and Component Selection

Sensor selection was guided by the operating environment of a commercial piggery: high ambient humidity, a corrosive ammonia atmosphere, mechanical vibration from pig activity, and the need for low-cost scalability across multiple feeder nodes.

Fig. 8. HX711 & Straight Bar Load Cell

- **Feed Weight (Load Cell + HX711):** A strain-gauge load cell rated for the expected trough load is interfaced to an HX711 24-bit sigma-delta ADC. The HX711 communicates with the ESP32 over a dedicated two-wire serial interface, providing 10 or 80 samples per second. The trough weight is logged before and after each feeding event; the delta constitutes the dispensed mass. Consumption is derived by tracking weight loss between feeding cycles.

Fig. 9. JSN-SR04T Waterproof Ultrasonic Sensor

- **Water Availability (Level Sensor):** A waterproof ultrasonic sensor is mounted above each pen's water trough. The ESP32 triggers an ultrasonic pulse and measures the echo time-of-flight to calculate the distance to the water surface. This determines the current water volume and triggers the solenoid valve to refill the trough when levels fall below the required baseline, ensuring constant water availability.

Fig. 10. MQ-135 Air Quality Sensor

- **Air Quality (MQ-135):** The MQ-135 electrochemical sensor detects ammonia (NH_3), CO_2, and benzene vapors—the primary respiratory hazards in piggery environments arising from urine decomposition. The sensor's analog output is read by the ESP32's onboard ADC and compared against safe-range thresholds; exceedances trigger caretaker alerts on the dashboard.

G. Wired Communication: RS-485 / MODBUS RTU

The system specifies a hardwired network topology to eliminate dependence on wireless signal integrity in metal-framed piggery structures. RS-485 with the MODBUS RTU protocol was selected as the communication backbone for the following reasons:

- Differential signaling on a twisted-pair bus provides high noise immunity in electrically harsh farm environments.
- The standard supports up to 32 nodes on a single bus segment at distances of up to 1,200 m, enabling future expansion of feeder modules without architectural redesign.
- MODBUS RTU is an industry-standard protocol with extensive ESP32 and Raspberry Pi library support, reducing firmware development overhead.

Each feeder-node ESP32 is interfaced to the bus via a MAX485 half-duplex transceiver IC. The Raspberry Pi master polls each slave node at configurable intervals, retrieves sensor data packets, and writes dispensing commands. A unique MODBUS device address is assigned to each feeder module during commissioning.

H. Application Layer and Data Persistence

The centralized user interface (UI) is hosted on the Raspberry Pi and rendered through a 7-inch DSI touchscreen mounted on the main control module enclosure. The software stack utilizes a lightweight web-based architecture to ensure low-latency updates:

- **Backend Stack:** A Flask-based web server manages the API endpoints for the dashboard. It interfaces with an SQLite database for ACID-compliant persistence of timestamped consumption logs, feeding schedules, and air-quality telemetry.
- **Frontend Stack:** A Chromium kiosk browser displays a dashboard built with responsive numeric gauges and time-series charts for real-time monitoring and historical trend visualization. It includes configuration panels for feeding schedules and alert logs for threshold exceedances.
- **Fault Tolerance:** All data is stored locally, ensuring continuous operation during internet outages, which is a primary concern identified in the study's scope and limitations. To address potential grid instability, a UPS module with 18650 lithium cells provides bridge power, ensuring that the SQLite database maintains integrity during short-duration power interruptions.
- **LoRa Alert Gateway:** A background Python thread runs alongside the Flask backend. It continuously monitors the SQLite database for newly generated threshold exceedances. To optimize radio duty cycles and prevent packet collisions, the gateway implements a message-queuing protocol, bundling alerts and transmitting them to the remote terminal at configured intervals or triggering immediate transmission for critical air-quality hazards.

I. Consumption Baseline and Anomaly Detection

A pig's health status is closely linked to its feeding and drinking behavior; measurable deviations in dietary intake are often among the first observable indicators of illness [**?**], [**?**]. The system establishes per-module baselines using a rolling time-window approach consistent with data-driven livestock monitoring methods reported in the literature [**?**], [**?**]. Let C_i denote the feed consumed in session i. The rolling mean over a window of N sessions is:

$$\mu_N = \frac{1}{N} \sum_{i=t-N+1}^{t} C_i \tag{7}$$

A deviation alert is generated when the current session's consumption C_t falls outside the band $[\mu_N - k\sigma, \mu_N + k\sigma]$, where σ is the rolling standard deviation and k is a configurable sensitivity coefficient (default $k = 2$). This statistical approach reduces false positives arising from natural day-to-day variation while reliably flagging behaviorally significant declines [?], [?].

J. Evaluation Metrics and Verification Protocol

System performance will be assessed against quantitative metrics aligned with the project's specific objectives. These metrics are intended to reflect the reliability of dispensing, sensing, communication, and health-monitoring functions commonly emphasized in precision livestock and embedded monitoring studies [?], [?], [?]. To ensure system reliability and accuracy prior to full-scale deployment, a rigorous testing protocol will be executed to systematically validate these metrics.

Quantitative Metrics:

- **Feed Dispensing Accuracy (FDA):** Percentage agreement between the target dispensed mass and the gravimetrically measured actual dispensed mass across a calibration trial of 30 dispensing events per module. A minimum passing threshold of $\geq 95\%$ is required for the system to be considered operationally accurate [?], [?]:

$$FDA = \left(1 - \frac{|M_{target} - M_{actual}|}{M_{target}}\right) \times 100\% \qquad (8)$$

- **Alert Response Accuracy (ARA):** Proportion of simulated consumption anomaly events correctly detected and flagged by the baseline algorithm within one polling cycle. A minimum passing threshold of $\geq 90\%$ is required to validate the reliability of the anomaly detection logic [?], [?]:

$$ARA = \frac{\text{True Positive Alerts}}{\text{Total Induced Events}} \times 100\% \qquad (9)$$

- **System Uptime** (U_{sys})**:** Measured over a 72-hour continuous operation trial, consistent with endurance validation practices for embedded monitoring systems [?], [?]:

$$U_{sys} = \frac{T_{total} - T_{downtime}}{T_{total}} \times 100\% \qquad (10)$$

  where T_{total} is the total trial duration in hours and $T_{downtime}$ is the cumulative time during which sensor polling or the dashboard was unavailable.
- **Data Display Latency:** Time elapsed between a sensor reading being acquired at the feeder node and its appearance on the dashboard, measured across 50 consecutive poll cycles. Target threshold: ≤ 2 seconds [?], [?].
- **Packet Delivery Ratio (PDR):** The percentage of LoRa alert payloads successfully received by the remote terminal compared to the total number of alerts transmitted by the central unit. A minimum passing threshold of $\geq 95\%$ is required to confirm reliable remote notification delivery [?], [?]:

$$PDR = \frac{\text{Total Packets Received}}{\text{Total Packets Transmitted}} \times 100\% \qquad (11)$$

- **Received Signal Strength Indicator (RSSI):** Measurement of the power present in the received LoRa radio signal, used to determine the maximum effective range of the notification system across the farm layout. An RSSI value of ≥ -120 dBm at the maximum tested distance is required to confirm an adequate communication link [?], [?].

K. Verification Phases

The testing protocol is divided into three primary phases to validate both the hardware edge nodes and the central software orchestration. This phased approach is consistent with common validation practice in embedded sensing, livestock-monitoring, and agricultural automation studies, where component calibration, communication validation, and full-system integration are assessed separately before deployment [?], [?], [?]:

- **Phase 1: Sensor Calibration and Hardware Validation**
  - **Gravimetric Testing (FDA Validation):** The load cell and HX711 assemblies will be calibrated using standard reference weights across a range of 0 to 5 kg, covering the expected trough load. A minimum of 10 calibration trials per load cell will be conducted. Linearity and hysteresis will be evaluated across the expected load range to ensure Feed Dispensing Accuracy meets the $\geq 95\%$ passing threshold [?], [?].
  - **Water Level Calibration:** The ultrasonic level sensors will be validated against manual depth measurements using a calibrated dipstick across a range of operational trough depths. A minimum of 10 trials per sensor will be conducted to establish a precise time-of-flight to depth conversion factor and verify the threshold-trigger logic for the solenoid valves [?], [?].
  - **Environmental Baseline:** The MQ-135 sensors will undergo a minimum 48-hour burn-in period followed by calibration in a controlled, clean-air environment to establish a baseline resistance (R_0) before exposure to the piggery atmosphere. Sensors will then be tested against known ammonia concentrations in the range of 10 to 300 ppm to verify threshold response accuracy prior to deployment [?], [?], [?].
- **Phase 2: Communication and Firmware Testing** The RS-485 wired bus will undergo signal integrity testing with all feeder nodes active on the bus simultaneously. A digital storage oscilloscope will be used to verify that differential voltage levels meet the RS-485 standard minimum of ± 1.5 V and ensure adequate common-mode noise rejection. MODBUS RTU packet loss rates will be quantified under simulated electrical noise conditions at the maximum prototype cable length. A packet loss rate of $\leq 1\%$ is required to confirm robust master-slave polling before proceeding to system integration [?], [?].
- **Phase 3: System-Level Integration (Uptime, Latency, ARA, and LoRa Validation)** A 72-hour continuous operation trial will be conducted in a controlled environment to measure System Uptime (U_{sys}) and verify

that Data Display Latency remains $\leq$ 2 seconds. To validate the rolling-window anomaly detection, simulated consumption drops will be manually injected into the data stream to test the Alert Response Accuracy (ARA) of the Python backend. Furthermore, the LoRa wireless link will undergo range and penetration testing. Simulated consumption drops will be injected to trigger alerts, and the RSSI and Packet Delivery Ratio (PDR) will be measured at varying distances from the main control module to validate the reliability of remote caretaker notifications [**?**], [**?**], [**?**].

L. Deployment Plan

The transition from the laboratory prototype to field operation in a commercial piggery will follow a structured, four-phase deployment strategy to minimize disruption to existing farm operations:

- **Phase 1: Site Assessment and Electrical Routing:** An evaluation of the piggery's physical layout will determine the optimal routing for the RS-485 twisted-pair cables and DC power distribution. Cable trajectories will be explicitly planned to avoid electromagnetic interference (EMI) from existing high-voltage farm equipment or large motors. This phase must produce three deliverables before installation proceeds: a cable routing diagram, a DC power distribution plan, and an EMI risk assessment identifying high-interference zones within the facility.
- **Phase 2: Physical Installation:** The central 0.18 m^3 hopper, auger mechanism, and main control module (Raspberry Pi) will be securely mounted at the centralized feeding station. Individual feeder nodes (ESP32 enclosures, load cells, and level sensors) will be mechanically integrated into the pen structures. The level sensors will be securely mounted directly above the water troughs, and the plumbing will be retrofitted to integrate the automated solenoid valves. Installation is considered complete only after all nodes pass a hardware checklist confirming that all units are powered, all sensors are returning non-zero readings, and all MODBUS device addresses have been successfully assigned and acknowledged by the master.
- **Phase 3: Commissioning and Baseline Initialization:** Once powered, unique MODBUS device addresses will be assigned to each feeder node. The system will initially operate in a "silent monitoring" phase (e.g., 7 to 14 days). During this period, the system will dispense feed and log data to the SQLite database without triggering health alerts, allowing the rolling-window algorithm to establish accurate, pen-specific consumption baselines (μ_N). Anomaly detection will not be activated until a minimum of $N = 14$ feeding sessions have been recorded per module, ensuring the rolling-window baseline is statistically sufficient before alerts are generated.
- **Phase 4: Caretaker Turnover and Training:** Farm operators will undergo practical training on utilizing the Raspberry Pi touchscreen interface. This includes instructing them on how to interpret the time-series charts, modify feeding schedules, address alert logs, and perform routine maintenance such as clearing auger jams or cleaning the optical/gas sensor interfaces. Turnover is considered complete only when each caretaker can independently demonstrate three competencies without assistance: interpreting and responding to a generated alert, modifying a feeding schedule on the dashboard, and performing basic sensor maintenance.

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